Topic: Section 1

1. In Part 2, what resistor value did you use in your circuit? Why did you choose this specific value?

   Using $V = IR$, finding a source that noted the forward voltage and operating current for the LED color I was able to find the needed resistor to meet the operating current.Then I rounded to the closest resistor I had R=250

2. Why was that resistor necessary? What does this imply about the internal resistance of the LED?

   The LED needs help limiting the current passing through. This means that the LED can pull enough current to damage itself and blow up.

3. Arduino uses the binary variables HIGH and LOW for digital pins. When you use digitalRead() on the Arduino Uno, above what voltage do readings end up being evaluated as HIGH and below what voltage are they LOW? Find the specific page in the online Arduino documentation that provides this information and link to it in your answer. Hint 1: The answers are not 0 or 5V. In reality, there are a range of voltages that are mapped to these readings.

   low $\leq$ 1.5V and high $\geq$ 3.0V

   https://docs.arduino.cc/language-reference/en/variables/constants/highLow/

Topic: Section 2

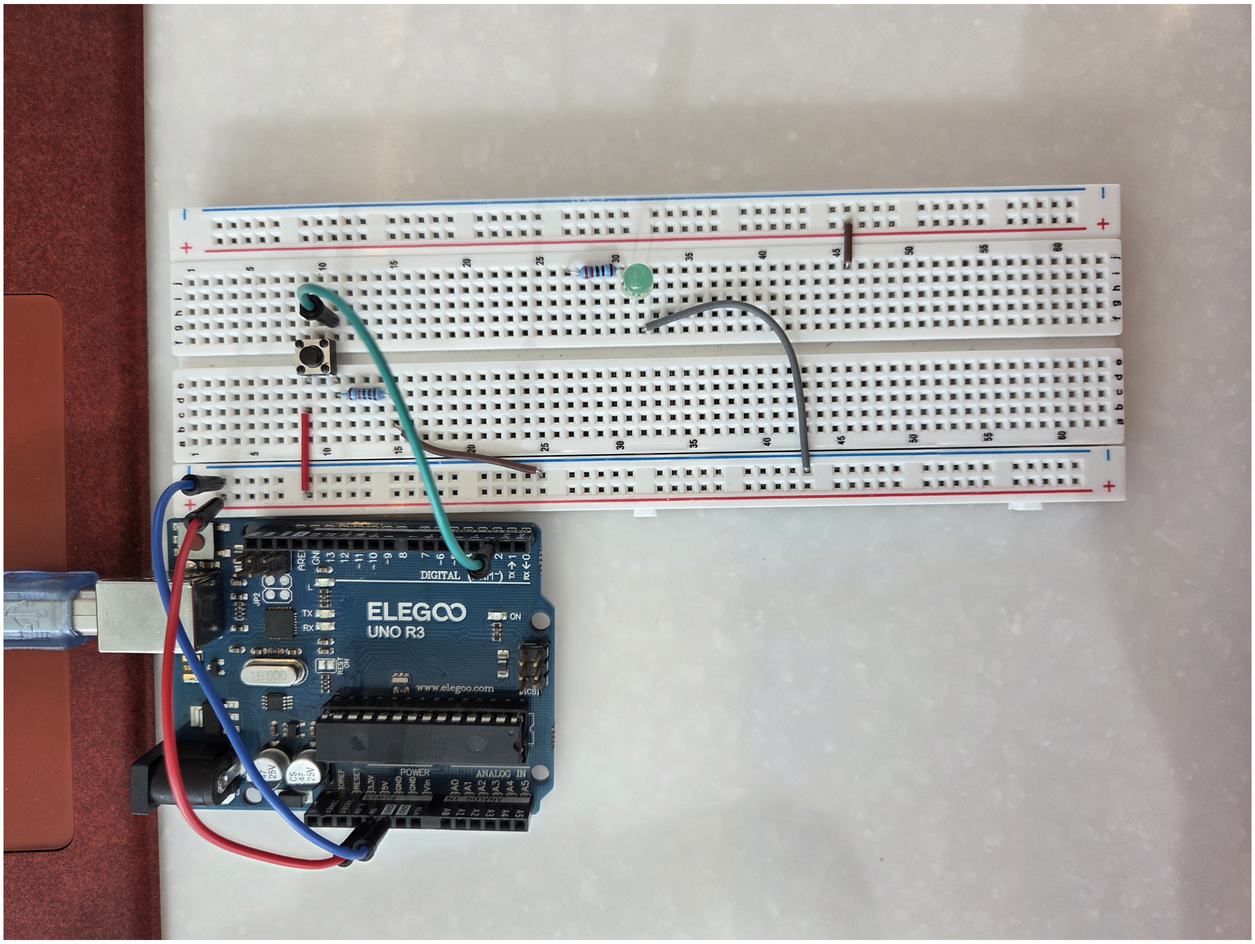

Figure 1: Pull-Up

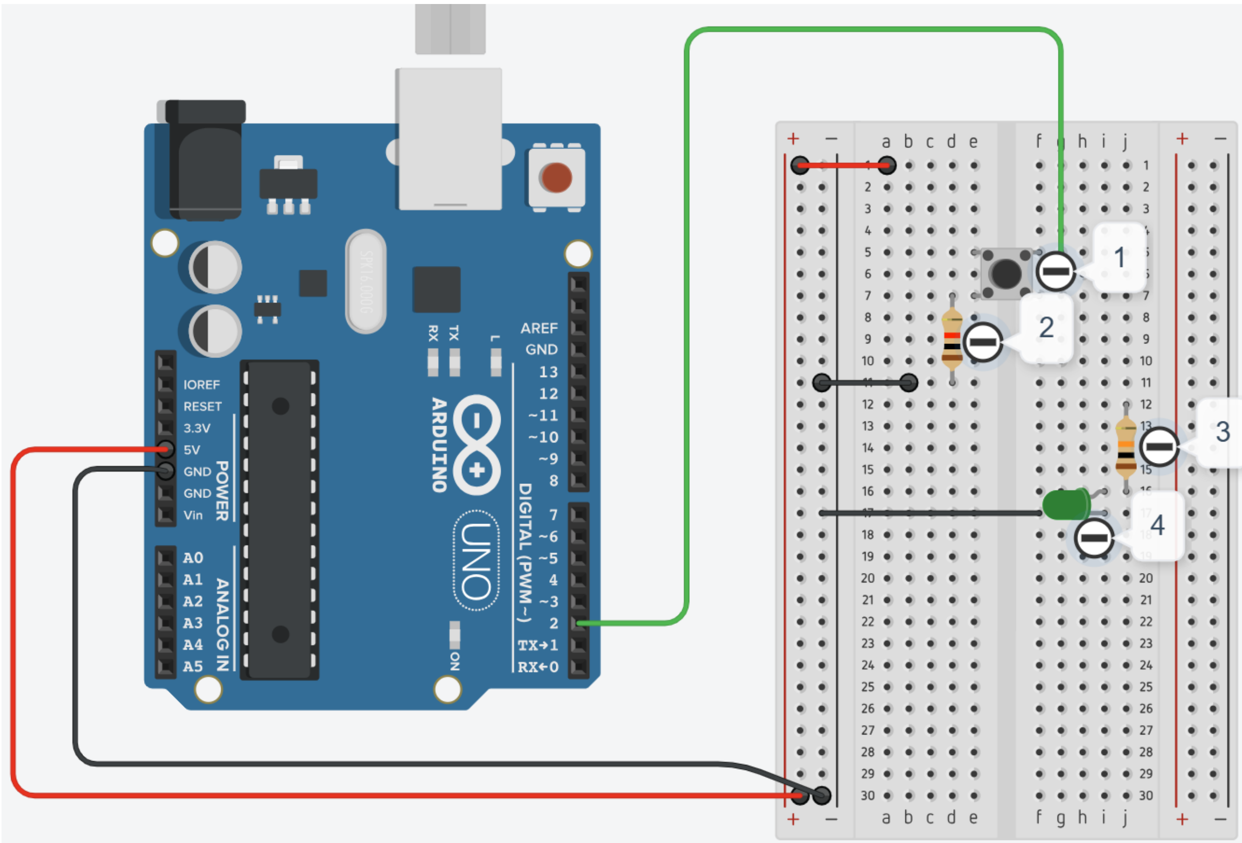

Figure 2: TinkerCAD model of sign off 1

Component Number	Detail
1	Button
2	$R_1 = 10k$
3	$R_2 = 220$
4	LED

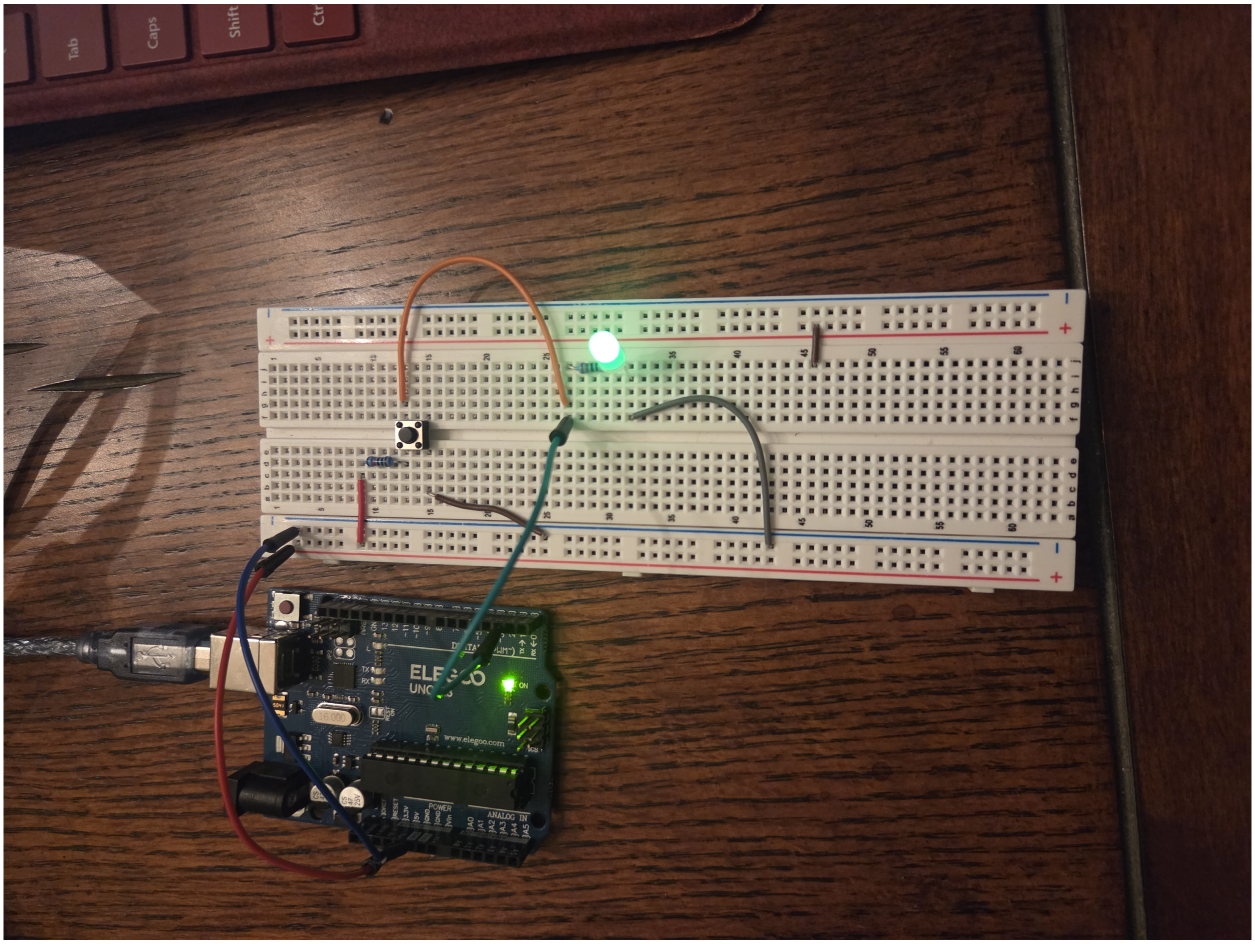

Figure 3: Sign off 2 Pull-Down

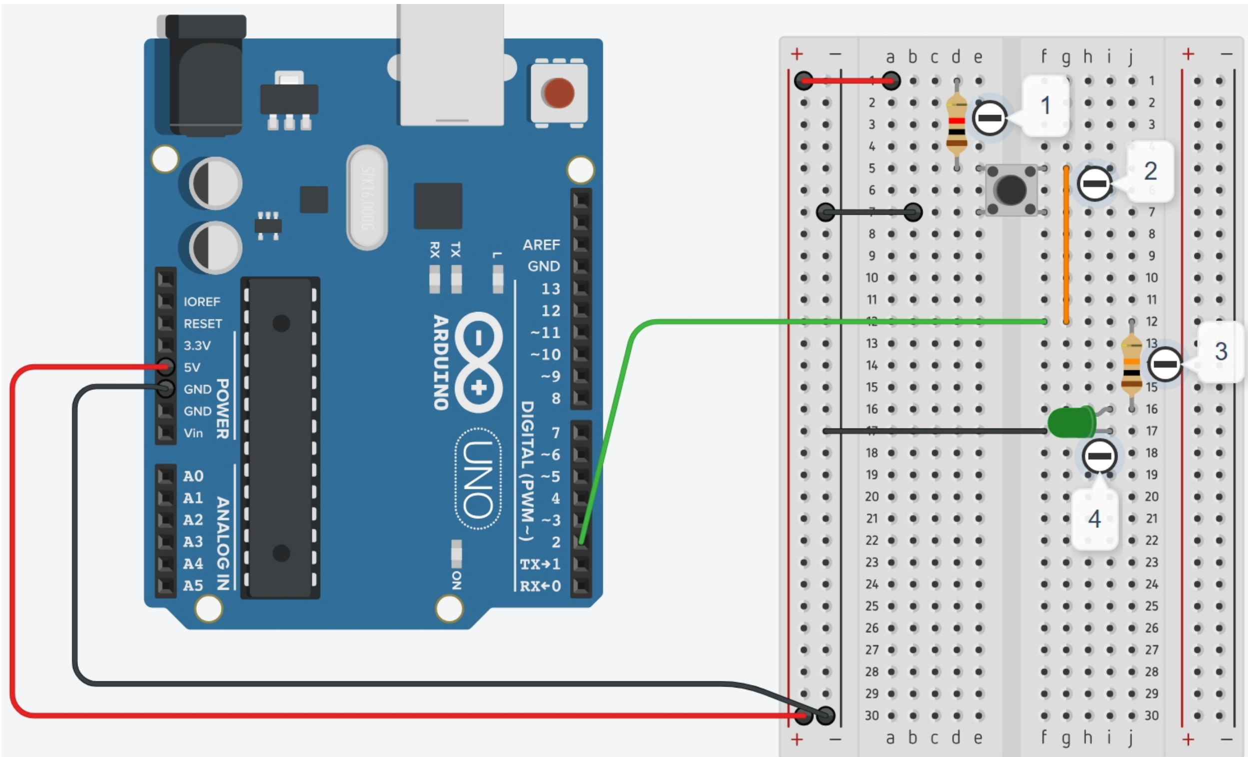

Figure 4: TinkerCAD model of sign off 2

Component Number	Detail
1	$R_1 = 10k$
2	Button
3	$R_2 = 220$
4	LED

You would want a Pull Up config when you are trying to do a type of emergency stop. Itll be on until you press it. A Pull Down would be good in a spot where you are wanting to control flow, maybe like a valve for your wall mounted cereal dispenser.

Topic: Section 3

https://forum.arduino.cc/t/turn-on-led-builtin-and-another-led-with-single-digitalwrite-command/1019550/18
https://deepbluembedded.com/arduino-analog-pins-as-digital-output-input/
https://www.petervis.com/electronics/led/led-resistor-calculator.html
Did not work with a student.

Topic: Section 4

1. What did you expect to learn during this lab? (Please respond in under 100 words.)

I expected to learn about how to change a pulldown to a pull up by changing hardware.

2. How would you rate your participation in class during this lab experience? Are any adjustments needed in your approach for future labs, in order for you to be successful? (Please respond in under 100 words.)

This one was pretty easy to get a handle on but I am taking the prelabs seriously for the next ones.

3. How well did you do at the lab assignment itself? Please assign yourself a grade out of four possible points, and explain your grade assignment using the rubric (see the Lab 1 assignment page on Canvas). You might also include thinking about strengths and weaknesses of your performance in this response. (Please respond in under 100 words.)

I would give myself a 4/4, I do find it hard to make an easy workflow to handle the labling but all the other things are easy to get done

4. What did you actually learn in the lab? (This might include a range of things, from subject matter and concrete skills to ways of knowing and working together with peers.) (Please respond in under 100 words.)

I learned that I struggle with physically changing the circuit to match the diagrams I draw to get what I am being asked for.

Topic: Questions to Think About

These questions used to be a component of the lab reports and existed for students to practice searching, reading, and critical thinking skills.

Consider the questions below as part of your preparation for the final exam; they are no longer part of the lab assignment, but they are fair game for later knowledge assessments.
Talk to the teaching staff as needed to check your understanding.

1. **Grounding**
   (a) What does ground on the Arduino represent in terms of voltage?
   0 [V]
   (b) In electronics, what is a common ground? When and why is it necessary?
   A common ground is a shared reference voltage (usually 0V) used across multiple circuits to ensure consistent and safe operation. We need it to be able to take mesurements of potential differences.
   (c) Is the Arduino ground or the common ground the same as Earth ground? Why or why not?
   They can be the same but they are not alwyas the same. Arduino ground is local to the circuit and may float unless tied to Earth ground.
2. **Polarity**
   (a) What is electronic polarity? What do we have to be careful about for polarized components?
   Electronic polarity is the direction that the current flows. Not paying attention to this can cause damage.
   (b) What component(s) from this lab are polarized?
   the LED is a polarized component.
3. **Digital Reading and Writing**
   (a) Which pins on the Arduino Uno can you use `digitalRead()` and `digitalWrite()` on?
   Digital pins 0-13 can use `digitalRead()` and `digitalWrite()` but I can also use the A0-A5 pins if I absolutley need to with some code.

```
void setup() {
   pinMode(A0, OUTPUT);
}
void loop() {
   digitalWrite(A0, HIGH);
   delay(1000);
   digitalWrite(A0, LOW);
   delay(1000);
}
```

Figure 5: code to use A pins as digital

(b) When you use `digitalRead()` on the Arduino Uno, above what voltage do readings end up being evaluated as HIGH and below what voltage are they LOW?
A high signal need pin [V] $>$ 3V
A low signal needs pin [V] $<$ 1.5 V

HIGH

The meaning of `HIGH` (in reference to a pin) is somewhat different depending on whether a pin is set to an `INPUT` or `OUTPUT`. When a pin is configured as an `INPUT` with `pinMode()`, and read with `digitalRead()`, the Arduino board (ATmega) will report `HIGH` if:

- a voltage greater than 3.0V is present at the pin (5V boards)
- a voltage greater than 2.0V is present at the pin (3.3V boards)

Figure 6: High

LOW

The meaning of LOW also has a different meaning depending on whether a pin is set to INPUT or OUTPUT. When a pin is configured as an INPUT with pinMode(), and read with digitalRead(), the Arduino board (ATmega) will report LOW if:

- a voltage less than 1.5V is present at the pin (5V boards)
- a voltage less than 1.0V (Approx) is present at the pin (3.3V boards)

Figure 7: Low

Code

```
%CODE HERE
```